Customer Shopping Behavior Analysis — Report

Python EDA • PostgreSQL • SQL • Power BI

November 2026

1. Project Overview

This project is an end-to-end analysis of retail customer shopping behavior. It starts from a raw transaction CSV, cleans and enriches it with Python (pandas), loads the clean table into PostgreSQL, answers ten business questions with SQL, and visualizes the results in a Power BI dashboard.

Pipeline

```
customer_shopping_behavior.csv
        |  (Python / pandas — cleaning + feature engineering)
        v
customer_shopping_behavior_analysis.ipynb
        |  (SQLAlchemy -> PostgreSQL, database: customer_behavior, table: customer)
        v
customer_behavior_sql_queries.sql   (10 business queries, Q1–Q10)
        |  (results captured as q1.png … q10.png)
        v
customer_behavior_dashboard.pbix    (Power BI dashboard)
```

Tools used: Python 3.9+, pandas, Jupyter Notebook, PostgreSQL (via SQLAlchemy/psycopg2), SQL, Power BI.

Files in this repo

File	Purpose
`customer_shopping_behavior.csv`	Raw dataset, 3,900 rows × 18 columns
`customer_shopping_behavior_analysis.ipynb`	Cleaning, feature engineering, PostgreSQL load
`customer_behavior_sql_queries.sql`	Q1–Q10 business queries on the `customer` table
`customer_behavior_dashboard.pbix`	Power BI dashboard
`q1.png` … `q10.png`	Screenshots of each query's result

2. Dataset Summary

- **Shape:** 3,900 rows × 18 columns. One row = one transaction.
- **Missing values:** only 37 nulls, all in `Review Rating` (later imputed).
- **No duplicates / type issues:** IDs run 1–3,900; numeric columns are clean.

Columns

Column	Type	Range / Values
Customer ID	int	1 – 3,900
Age	int	18 – 70 (mean 44.1)
Gender	category	Male (2,652), Female (1,248)
Item Purchased	category	25 products (Blouse, Pants, Jewelry …)
Category	category	Clothing (1,737), Accessories (1,240), Footwear (599), Outerwear (324)
Purchase Amount (USD)	int	20 – 100 (mean 59.76, total revenue **$233,081**)
Location	category	50 US states
Size	category	S, M, L, XL
Color	category	25 colors
Season	category	Spring (999), Fall (975), Winter (971), Summer (955) — almost even
Review Rating	float	2.5 – 5.0 (mean 3.75; 37 nulls)

Column	Type	Range / Values
Subscription Status	Yes/No	No (2,847), Yes (1,053)
Shipping Type	category	6 types, ~627–675 orders each — almost even
Discount Applied	Yes/No	No (2,223), Yes (1,677 → 43% of orders)
Promo Code Used	Yes/No	Identical to Discount Applied on every row (redundant)
Previous Purchases	int	1 – 50 (mean 25.4)
Payment Method	category	6 methods, ~612–677 orders each — almost even
Frequency of Purchases	category	7 frequencies, ~539–584 orders each — almost even

Key observations from the raw data

- Sales are evenly spread across seasons, states, shipping types, payment methods and

purchase frequencies — no single channel dominates, which is good for analysis (comparisons are fair).

- Average order value is flat (~$57–$60) across all four categories and both genders,

so revenue differences come from **volume**, not price.

- 43% of orders used a discount — large enough to measure discount impact reliably.
- `Promo Code Used` mirrors `Discount Applied` on all 3,900 rows, so it carries no

extra information.

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3. Exploratory Data Analysis Using Python

The notebook (`customer_shopping_behavior_analysis.ipynb`) works through these stages:

3.1 Loading and inspection

- Load the CSV with `pd.read_csv`.
- `head()`, `info()`, `describe(include='all')` to confirm 3,900 × 18, check dtypes

(numeric: Customer ID, Age, Purchase Amount, Review Rating, Previous Purchases; rest categorical) and value ranges.

3.2 Missing-value treatment

- `isnull().sum()` finds exactly **37 missing Review Ratings** and nothing else.
- Fix: fill each null with the **median rating of its own category**

(`groupby('Category').transform(lambda x: x.fillna(x.median()))`), which preserves category-level differences instead of flattening everything toward one global value.

- Re-check confirms zero nulls.

3.3 Column-name normalisation

- Lowercase + underscores, and `purchase_amount_(usd)` → `purchase_amount`, so the

cleaned frame has database-friendly snake_case names.

3.4 Feature engineering

- `age_group` — age split into quartiles with `pd.qcut`, giving four balanced

segments of ~940–1,030 customers each: Young_Adult (18–31), Adult (31–44), Middle-aged (44–57), Senior (57–70).

- `purchase_frequency_days` — the 7 text frequencies mapped to days

(Weekly→7, Fortnightly/Bi-Weekly→14, Monthly→30, Quarterly/Every 3 Months→90, Annually→365) so buying frequency becomes numeric and comparable.

3.5 Redundancy removal

- Verified `(discount_applied == promo_code_used).all()` is `True`, then dropped `promo_code_used` — one fewer column, zero information lost.

3.6 Loading into PostgreSQL

- SQLAlchemy engine to a local `customer_behavior` database, `df.to_sql('customer', if_exists='replace')` — the cleaned table that every SQL query in Section 4 runs on.

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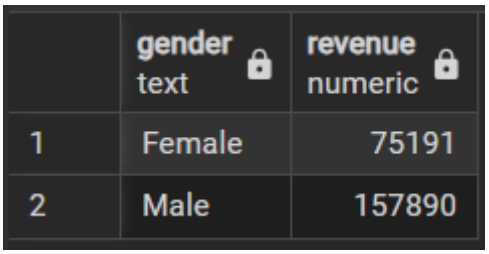
4. Data Analysis Using SQL (Business Questions)

All queries run on the `customer` table in the `customer_behavior` database. Total revenue across every query reconciles to **$233,081**.

Q1. Total revenue: male vs. female customers

Gender	Revenue
Male	157,890
Female	75,191

Insight: Male customers generate ~68% of revenue — but that mirrors their ~68%



	gender text	revenue numeric
1	Female	75191
2	Male	157890

share of transactions (2,652 of 3,900). Average spend per order is nearly identical ($59.5 male vs. $60.3 female), so there is no gender pricing/spend gap; the revenue gap is purely footfall.

Q2. Discount users who still spent above the average order ($59.76)

839 of the 1,677 discount orders (about half) are at or above the overall average purchase amount — e.g. customer 43 spent $100 with a discount applied.

Insight: Discounts are not confined to cheap baskets; high-value customers use them

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62
11	24	88
12	29	94
13	32	79
14	33	67
15	35	91
16	37	69
17	40	60
18	41	76
19	43	100
20	44	69
21	55	94
22	57	73

too, so discount strategy reaches the customers who matter most.

Q3. Top 5 products by average review rating

Product	Avg rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.79

Insight: Ratings cluster tightly (3.79–3.86) — no runaway winner and no problem

	item_purchased text	Average Product Rating numeric
1	Gloves	3.86
2	Sandals	3.84
3	Boots	3.82
4	Hat	3.80
5	Skirt	3.78

product, with footwear/gloves slightly ahead. Quality looks consistent.

Q4. Average order: Standard vs. Express shipping

Shipping	Avg order
Standard	58.46
Express	60.48

Insight: Express buyers spend ~$2 more per order — customers paying for speed also

	shipping_type text	avg numeric
1	Standard	58.4602446483180428
2	Express	60.4752321981424149

buy slightly bigger baskets.

Q5. Do subscribed customers spend more?

Subscription	Customers	Avg spend	Total revenue
No	2,847	59.87	170,436
Yes	1,053	59.49	62,645

Insight: No — average spend is virtually identical ($59.87 vs. $59.49). Subscribers

	subscription_status text	total_customer bigint	round numeric	sum numeric
1	No	2847	59.87	170436
2	Yes	1053	59.49	62645

are only 27% of customers and contribute 27% of revenue, so the subscription program currently drives reach, not higher baskets.

Q6. Top 5 products by discount usage rate

Product	Discount rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Insight: Roughly every second Hat/Sneakers/Coat order uses a discount — these

	item_purchased text	discount_rate numeric
1	Hat	50.00
2	Sneakers	49.00
3	Coat	49.00
4	Sweater	48.00
5	Pants	47.00

categories are the most promotion-sensitive and the best candidates for targeted offers.

Q7. Customer segmentation by previous purchases

Segment	Definition	Customers
Loyal	> 10 previous purchases	3,116 (~80%)
Returning	2 – 10	701 (~18%)
New	1	83 (~2%)

Insight: The base is overwhelmingly loyal — retention is strong, but only 2% are

	customer_segment text	Number of Customers bigint
1	Loyal	3116
2	New	83
3	Returning	701

genuinely new, so acquisition (not retention) is the growth bottleneck.

Q8. Top 3 products by orders within each category

Category	#1	#2	#3
Accessories	Jewelry (171)	Belt (161)	Sunglasses (161)
Clothing	Blouse (171)	Pants (171)	Shirt (169)
Footwear	Sandals (160)	Shoes (150)	Sneakers (145)
Outerwear	Jacket (163)	Coat (161)	— (only 2 products)

Insight: Clear hero products per category (Jewelry, Blouse/Pants, Sandals, Jacket).

	item_rank bigint	category text	item_purchased text	total_orders bigint
1	1	Accessori...	Jewelry	171
2	2	Accessori...	Sunglasses	161
3	3	Accessori...	Belt	161
4	1	Clothing	Blouse	171
5	2	Clothing	Pants	171
6	3	Clothing	Shirt	169
7	1	Footwear	Sandals	160
8	2	Footwear	Shoes	150
9	3	Footwear	Sneakers	145
10	1	Outerwear	Jacket	163
11	2	Outerwear	Coat	161

Outerwear has just two products (Jacket, Coat) — the thinnest assortment and a candidate for range expansion.

Q9. Are repeat buyers (> 5 previous purchases) likely to subscribe?

Subscription	Repeat buyers
No	2,518
Yes	958

Insight: No — of 3,476 repeat buyers, only ~28% subscribe, almost the same as the

	subscription_status text	repeat_buyers bigint
1	No	2518
2	Yes	958

overall 27% subscription rate. Buying often does not convert customers into subscribers, so the subscription pitch needs its own incentive.

Q10. Revenue contribution by age group

Age group	Revenue
Young_Adult (18–31)	62,143
Middle-aged (44–57)	59,197
Adult (31–44)	55,978
Senior (57–70)	55,763

Insight: Young adults lead revenue (~27%) despite similar group sizes — the

	age_group text	total_revenue numeric
1	Young_Adult	62143
2	Middle-aged	59197
3	Adult	55978
4	Senior	55763

highest value-per-customer segment and the one to prioritise in marketing.

Key Takeaways

1. **Volume, not price, drives revenue** — order values are flat across categories, genders and subscriber status.
2. **Loyal base, weak acquisition** — 80% loyal vs. 2% new customers.
3. **Subscription doesn't lift baskets** — subscribers spend the same per order as everyone else, and even heavy repeat buyers rarely subscribe.
4. **Discounts reach high spenders** — half of discounted orders are above-average value; Hat/Sneakers/Coat are the most promotion-sensitive products.
5. **Young adults are the top segment** — highest revenue of the four age groups.